

Week 1: Unsupervised Learning for Process Data

with the Dow Data Challenge

Unsupervised Learning in Chemical Engineering

- No labels, no targets — just raw process data
- Goals: discover **structure**, identify **patterns**, reduce **complexity**
- Common applications:
 - Fault detection and diagnosis
 - Process monitoring
 - Grouping similar operating conditions
 - Reducing analysis dimensions

Key question: What can the data tell us without being told what to look for?

The Dow Data Challenge Problem

- Industrial dataset from a **continuous chemical process** (Dow Chemical)
- ~40 process variables (temperatures, pressures, flows, compositions)
- Multiple operational modes and transitions
- Known and unknown fault conditions

Qin et al., "Integration of process knowledge and statistical learning for the Dow data challenge problem," *Comp. & Chem. Eng.*, **153**, 107451 (2021)

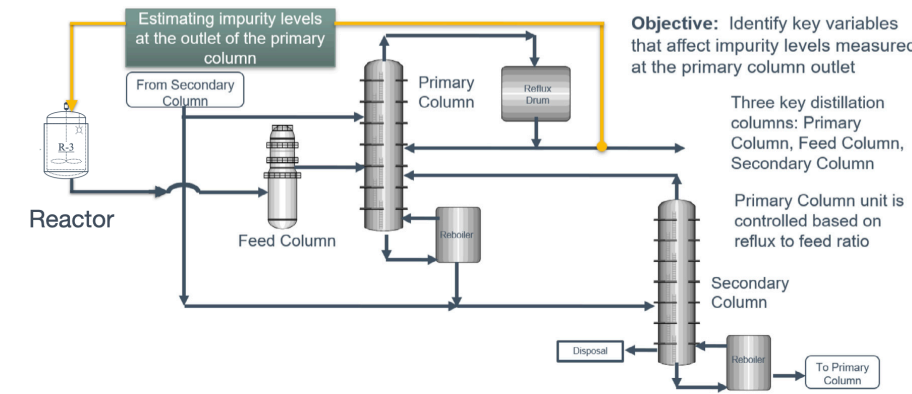
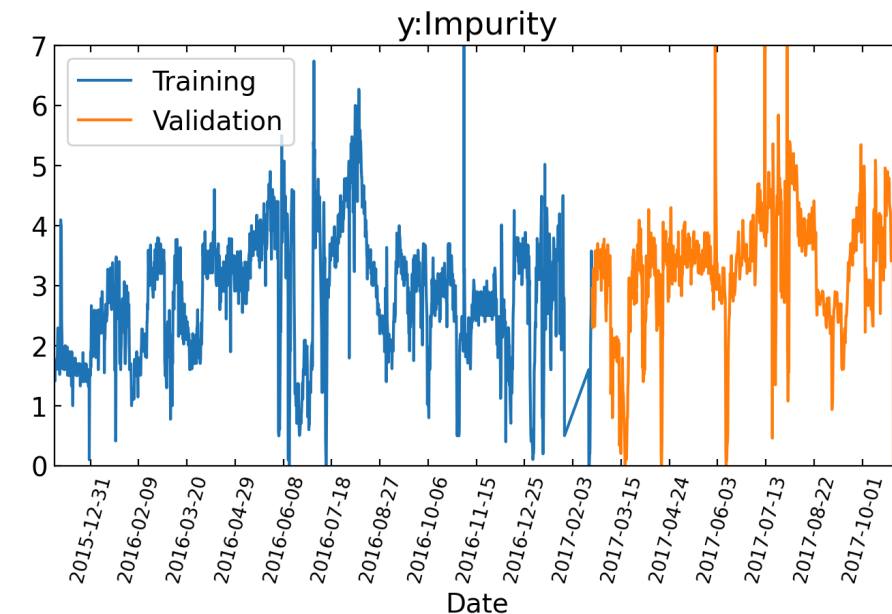


Fig. 1. Block diagram of the Dow data challenge problem considered for the industrial data analytics study.



The Dow Data Challenge Problem

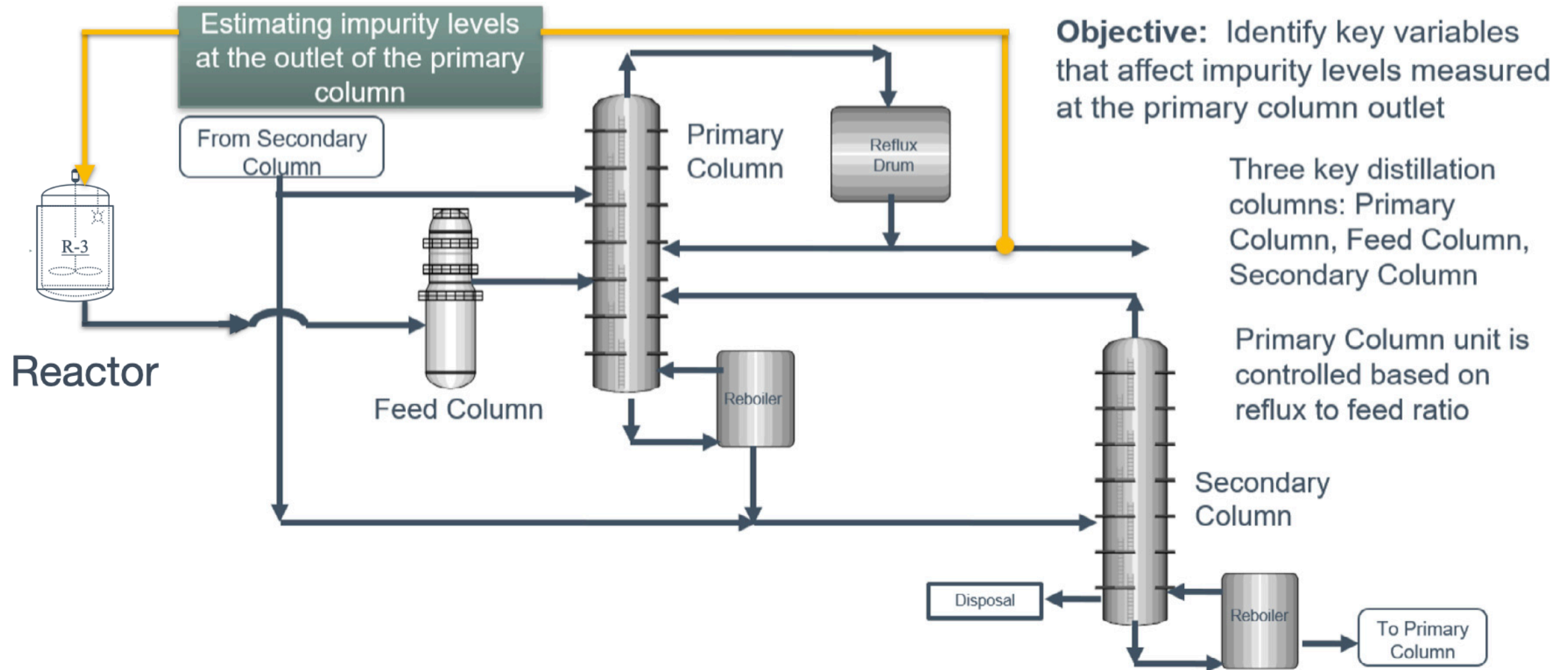


Fig. 1. Block diagram of the Dow data challenge problem considered for the industrial data analytics study.

Process sensors

Flow · Temperature · Pressure/DP · Level · Concentration

x1: Primary Column Reflux Flow
x2: Primary Column Tails Flow
x3: Input to Primary Column Bed 3 Flow
x4: Input to Primary Column Bed 2 Flow
x5: Primary Column Feed Flow from Feed Column
x6: Primary Column Make Flow
x7: Primary Column Base Level
x8: Primary Column Reflux Drum Pressure
x9: Primary Column Condenser Reflux Drum Level
x10: Primary Column Bed1 DP
x11: Primary Column Bed2 DP
x12: Primary Column Bed3 DP
x13: Primary Column Bed4 DP
x14: Primary Column Base Pressure
x15: Primary Column Head Pressure
x16: Primary Column Tails Temperature
x17: Primary Column Tails Temperature 1
x18: Primary Column Bed 4 Temperature
x19: Primary Column Bed 3 Temperature

x20: Primary Column Bed 2 Temperature
x21: Primary Column Bed 1 Temperature
x22: Secondary Column Base Concentration
x23: Flow from Input to Secondary Column
x24: Secondary Column Tails Flow
x25: Secondary Column Tray DP
x26: Secondary Column Head Pressure
x27: Secondary Column Base Pressure
x28: Secondary Column Base Temperature
x29: Secondary Column Tray 3 Temperature
x30: Secondary Column Bed 1 Temperature
x31: Secondary Column Bed 2 Temperature
x32: Secondary Column Tray 2 Temperature
x33: Secondary Column Tray 1 Temperature
x34: Secondary Column Tails Temperature
x35: Secondary Column Tails Concentration
x36: Feed Column Recycle Flow
x37: Feed Column Tails Flow to Primary Column
x38: Feed Column Calculated DP
x39: Feed Column Steam Flow
x40: Feed Column Tails Flow

y: Impurity

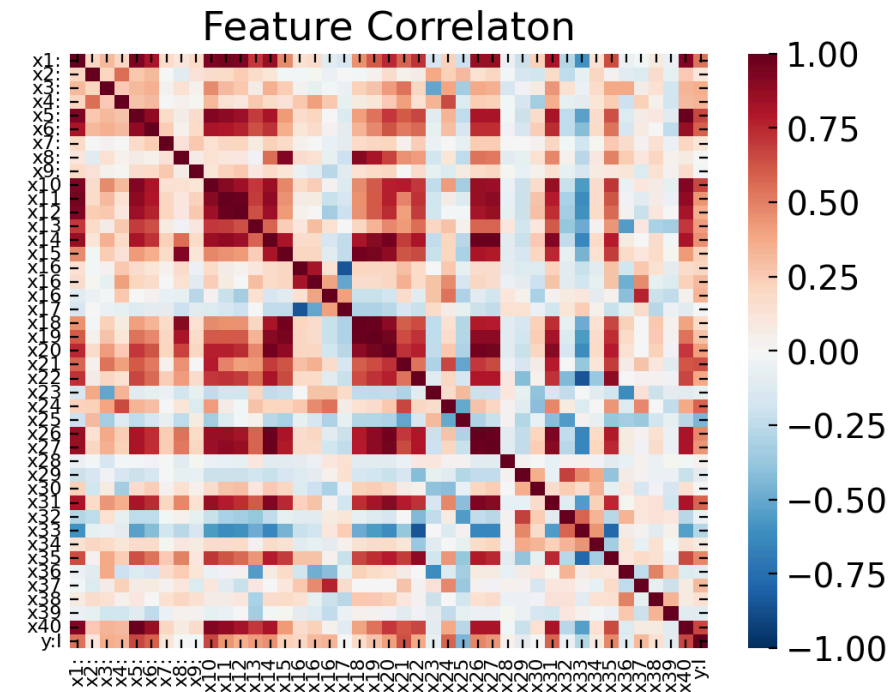
The Challenge

1. Use **unsupervised methods** to

- Identify distinct operational regimes
- Detect anomalous behavior
- Reduce dimensionality for visualization and modeling

2. Use **supervised methods** to

- Perform variable selection
- Account for process drift
- Predict impurity levels
- Analyze root causes

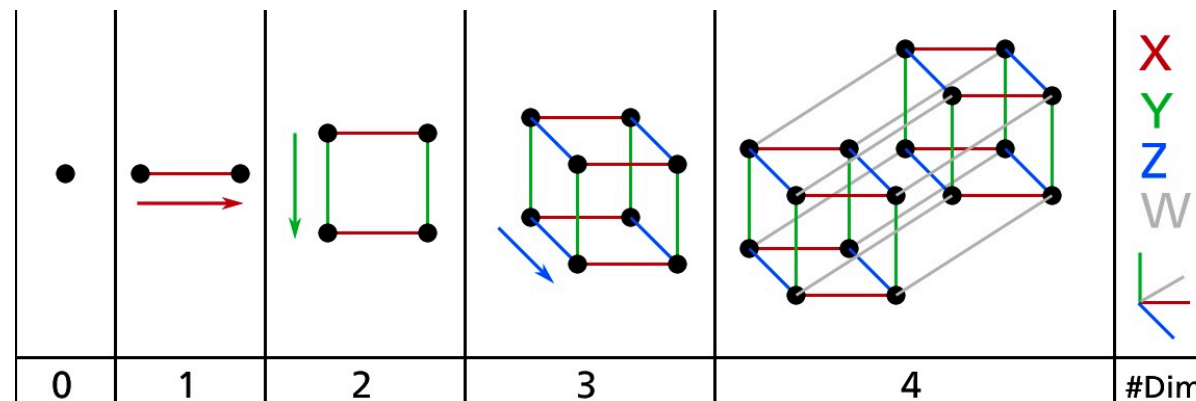


Dimensionality Reduction: The Curse of Dimensionality

~40 variables in the Dow dataset:

- Volume of the space **grows exponentially** with dimensions
- Data becomes **sparse** — need exponentially more samples to maintain density
- Models **overfit** more easily
- Many variables are **redundant** (correlated sensors, redundant measurements)

Goal: Project data onto a lower-dimensional subspace that captures the most important variation



Principal Component Analysis (PCA)

Find the directions of **maximum variance** in the data

- Transforms correlated variables into a smaller set of **uncorrelated** components
- Each component is a weighted combination of the original variables
- First PC captures the most variance, second PC the next most, and so on
- Choose enough components to retain most of the variance (e.g., 90–95%)

PCA is **linear** — it finds the best linear subspace, but cannot capture nonlinear structure

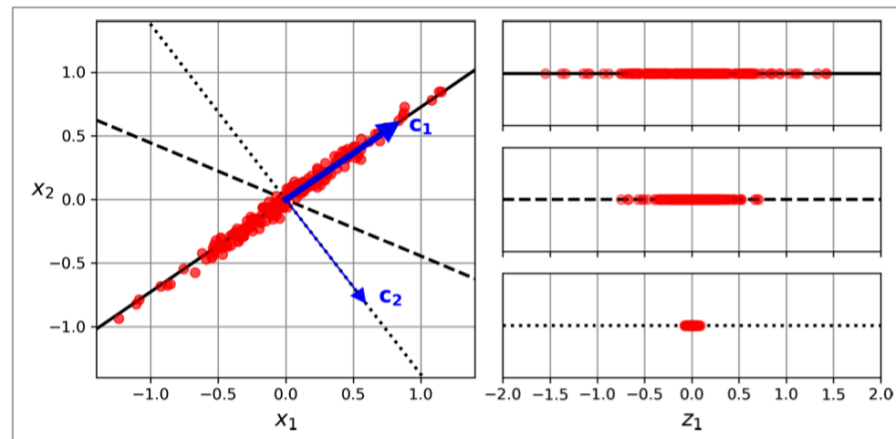


Figure 8-7. Selecting the subspace on which to project

PCA: The Math

1. Standardize the data matrix $\mathbf{X}$ ($n \times p$)
2. Compute the covariance matrix $\mathbf{C} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$
3. Eigendecomposition: $\mathbf{C} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$
 - $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots)$ — **eigenvalues** = variance captured by each component
 - $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots]$ — **eigenvectors** = the principal component directions
4. Keep the top k eigenvectors: $\mathbf{V}_k = [\mathbf{v}_1, \dots, \mathbf{v}_k]$ (**loadings**)
5. Project: $\mathbf{T} = \mathbf{X} \mathbf{V}_k$ — **scores** in the reduced space

1. J. Shlens, "A Tutorial on Principal Component Analysis", (2014). doi:10.48550/arXiv.1404.1100

2. S. Wold and K. Esbensen. P. Geladi, "Principal Component Analysis", (n.d.). doi:10.1016/0169-7439(87)80084-9

PCA for Process Monitoring

PCA defines a "normal operating envelope" in reduced space

Monitoring statistics:

- Hotelling's T^2 — variation *within* the PCA model (score space)
 - $T^2 = \mathbf{t}^T \mathbf{\Lambda}_k^{-1} \mathbf{t}$ High T^2 : unusual *but explainable* by the model
- Q statistic (SPE) — variation *outside* the model (residual space)
 - $Q = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 = \|\mathbf{x} - \mathbf{V}_k \mathbf{t}\|^2$
 - High Q : something new is happening that the model doesn't capture

1. T. Kourti and J. F. MacGregor, "Process analysis, monitoring and diagnosis, using multivariate projection methods", (n.d.). doi:10.1016/0169-7439(95)80036-9

Clustering: Identifying Operational Regimes

K-Means Clustering `sklearn.cluster.KMeans`

- Partition n observations into k clusters by minimizing within-cluster variance
- Objective: $\min_S \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$
- Fast, scalable — but assumes spherical clusters of similar size

Hierarchical Clustering `scipy.cluster.hierarchy.dendrogram, linkage`

- Builds a tree (dendrogram) of nested clusters
- Agglomerative (bottom-up): start with each point as a cluster, merge closest pairs
- No need to pre-specify k — cut the dendrogram at the desired level
- Linkage criteria: single, complete, average, Ward's

Choosing the Number of Clusters

Elbow method

- Plot within-cluster sum of squares (WCSS) vs. k
- Look for the "elbow" where adding clusters gives diminishing returns

Silhouette analysis

- For each point: $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$
 - $a(i)$: mean distance to points in same cluster
 - $b(i)$: mean distance to nearest neighboring cluster
- $s(i) \in [-1, 1]$; higher is better

Domain knowledge matters — in process data, clusters should correspond to physically meaningful operating regimes

Clustering Applied to Process Data

In the Dow dataset, clustering reveals:

- **Steady-state regimes** — groups of similar operating conditions
- **Transitions** — points between clusters indicate mode changes
- **Outliers** — points far from any cluster center may indicate faults

Practical considerations:

- **Scaling is critical** — variables have different units and ranges; always standardize
- **Time structure** — process data is sequential; consider windowing or time-lagged features
- **Correlation** — many process variables are highly correlated, which can distort distance metrics